



Habib University

Course Title: Introduction to Probability and Statistics

Instructor's name: Aamir Hasan and Ummamah Shah

Class ID: MATH 310/EE 354 (L3 & L4 sections)

Examination: Final - Term

Exam Date: May 04, 2026

Total Marks: 75

Time/ Duration: 2 hours (120 minutes)

Name _____

ID _____

For examiner's use only

Question	CLO No	Points	Score
1	5	15	
2	4	15	
3	3	15	
4	4	15	
5	4	15	
		Total = 75	

Instructions:

- Arrive 10 minutes before the start of the exam. Same communicated earlier.
- **Mobile phones, smart watches and other electronic devices must be turned off and put away before the start of the exam. You are allowed to use a scientific calculator (Not the one on your smart phone/watch. Sharing of calculators is not allowed as well).**
- No food or drink, other than water, is allowed in the examination room.
- You are not allowed to communicate with other candidates.
- You are not allowed to leave your seat unless granted permission by an invigilator.
- **Assume where necessary and justify your assumptions.**
- You are required to fill all answers in the provided answer booklet.
- You are allowed to bring a cheat sheet. It must be a handwritten A4 sheet of paper. You may write on both sides of the sheet. Photocopied cheat sheets are not allowed. You have to write your names on the cheat sheet before the start of the exam.
- The standard normal table has been provided on page 2.
- The question paper must be returned at the end of exam along with answer script.
- Write your name and ID on the question paper before the start of the exam.
- Good luck!

	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4	0.99997	0.99997	0.99997	0.99997	0.99997	0.99997	0.99998	0.99998	0.99998	0.99998

Table 1: The standard normal table

Question No 1 [15 points]

A hospital uses an AI-assisted screening system to detect a rare heart condition in patients. Patients are categorized into three groups:

- **Low-risk (L):** 70% of patients
- **Medium-risk (M):** 20% of patients
- **High-risk (H):** 10% of patients

The actual probability of having the disease depends on the group:

- $P(D|L) = 0.01$
- $P(D|M) = 0.05$
- $P(D|H) = 0.20$

The AI system gives a positive result (+) with the following performance:

- If the patient has the disease, $P(+|D) = 0.95$.
- If the patient does not have the disease, $P(+|D^c) = 0.08$.

- (a) [3 points] Find the overall probability that a randomly selected patient has the disease, $P(D)$.
- (b) [3 points] Find the probability that a randomly selected patient tests positive, $P(+)$.
- (c) [4 points] If a patient tests positive, what is the probability that the patient actually has the disease, $P(D|+)$?
- (d) [5 points] A patient tests positive. What is the probability that the patient belongs to the high-risk group, $P(H|+)$?

Question No 2 [15 points]

K-Electric is analyzing uncertainty in short-term electricity demand using the following three different modelling approaches:

Model 1: Discrete Peak Demand Scenario (X')

- Engineers model extreme demand events using a Binomial process: $X \sim \text{Binomial Distribution } (n = 2, p = 0.5)$.
- This represents the number of simultaneous high-load events (such as industrial spikes).
- To convert this into a standardized demand deviation, define: $X' = 2X - 2$.

Model 2: Seasonal Variation (Y)

- Seasonal fluctuations in demand are modelled using a uniform distribution: $Y \sim \text{Uniform } (-1, 3)$.

Model 3: Random Noise (W)

- Short-term fluctuations are modelled using a normal random variable, $W \sim N(1, 4)$.

Combined Demand Model

The final demand deviation Z is constructed by randomly selecting one of the above models:

- $Z = [X'$ with probability $1/4$, Y with probability $1/2$ and W with probability $1/4]$
- Assume that,
 - The model selection is random and all variables are independent.

- (a) [5 points] Determine and plot the CDF for X' , Y and W .
- (b) [7 points] Determine and plot the CDF for the final demand deviation, Z .
- (c) [3 points] Compute probability, $P(-1 < Z \leq 2)$ using the CDF plot in part b.

Question No 3 [15 points]

A university decides to use an AI-based analytics system in their admission process to evaluate student performance based on a holistic entrance exam. The Random Variable $\mathbf{X}$, is the cumulative entrance exam score (out of 100) which can be modelled as approximately normally distributed, $\mathbf{X} \sim \mathbf{N}(68, 12^2)$. The university defines:

- Pass: score ≥ 50
- Merit: score ≥ 80

- (a) **[4 points]** Find the probability that a candidate scores between 60 and 75.
- (b) **[6 points]** Given that a candidate has passed ($X \geq 50$), what is the probability that the candidate earned Merit ($X \geq 80$)? Interpret your answer.
- (c) **[5 points]** The university wants to award Merit to only the top 8% of candidates. Find the minimum score M required for Merit such that $P(X \geq M) = 0.08$.

Question No 4 [15 points]

The following two parts are unrelated:

- (a) **[8 points]** The lifetime $\mathbf{X}$ of a light bulb is a random variable with

$$P(\mathbf{X} > t) = \frac{2}{2+t} \text{ for } t > 0.$$

Suppose three new light bulbs are installed at time $t = 0$. At time $t = 1$ all three light bulbs are still working. Find the probability that at least one light bulb is still working at time $t = 9$.

- (b) **[7 points]** The random variable $\mathbf{X}$ is uniformly distributed in the interval $[0, a]$. Suppose a is unknown, so we estimate a by the maximum value observed in n independent repetitions of the experiment; that is, we estimate a by $Y = \max\{X_1, X_2, \dots, X_n\}$. Find the probability, $P(Y \leq y)$.

Question No 5 [15 points]

The Probability Density Function (pdf) for random variable $\mathbf{X}$ is given below. Conditioned on $X = x$, conditional pdf $f_{Y|X}(y|x)$ is uniformly distributed between $0 < y < 3x$.

- (a) **[3 points]** Calculate and plot the Cumulative Distribution Function (CDF), $F_X(x)$.
- (b) **[6 points]** Calculate and plot (showing all the working and the applicable region) for the Joint Probability Density Function, $f_{X,Y}(x, y)$.
- (c) **[6 points]** Calculate and plot (showing all the working and the applicable region) the marginal Probability Density Function, $f_Y(y)$.

